Collaborative Discussion 1: Agent Based Systems

This discussion will last for 3 weeks, starting in Unit 1 and ending in Unit 3.

Discussion Topic: Agent Based Systems

Discuss what has led to the rise of agent-based systems and the benefits that this approach can offer to organisations. Discussion should include at least three references.

Initial Post

by Yasmin Almeida Ferrera - Monday, 3 August 2026, 2:56 PM
Number of replies: 2

Agent-Based Systems: Decoding Their Rise and Business Value

An AI agent can be described as a system that acts on behalf of a user or organisation by taking an active role in achieving a specific goal. Unlike traditional software systems that mainly follow predefined instructions, AI agents can analyse information, make decisions, and interact with their environment to complete tasks. Large language models (LLMs), such as ChatGPT, represented an important development towards this approach, as they allowed users to communicate objectives through natural language and receive assistance in completing tasks. However, despite their initial impact, early LLM-based systems also revealed several limitations, including hallucinations, outdated information, limited context understanding, and difficulties in producing consistently reliable outputs.

The rise of agent-based systems has been driven by recent developments aimed at overcoming these limitations. Techniques such as Retrieval-Augmented Generation (RAG), prompt engineering, fine-tuning approaches, and the integration of external tools have expanded the capabilities of AI agents by allowing them to access additional information and perform more complex tasks. Furthermore, multi-agent systems have increased the potential of this approach by enabling different agents to collaborate and divide responsibilities when solving complex problems (Anthropic, 2024). Agent-based modelling has also been applied to complex systems where interactions between individual agents can help analyse and understand wider patterns and outcomes (Nature Index, n.d.).

As Bonnmann highlights, the main challenge for organisations is no longer whether AI agents will become valuable, but how quickly organisations can create the right conditions

for their adoption (Bonnmann, 2022). This reflects the increasing interest in using artificial intelligence not only for automation, but also as a tool to support decision-making and problem-solving.

For organisations, agent-based systems offer several benefits, including increased efficiency, scalability, and the automation of complex decision-making processes. A practical example can be seen in AI-powered investment platforms such as Wealthfront, which use automated agents to create personalised investment strategies based on an individual's preferences, risk tolerance, and financial objectives. Instead of simply selecting investments with the highest potential return, the system evaluates factors such as market conditions, diversification, and risk management to identify a portfolio that maximises long-term value for the user (Domo, 2026).

However, organisations must also consider the limitations of these systems. Issues such as hallucinations, biased outputs, and unreliable decisions demonstrate that human evaluation and supervision remain essential (Massachusetts Institute of Technology Media Lab, 2025). Therefore, the value of agent-based systems lies not in replacing human expertise, but in supporting human decision-making and enabling organisations to operate more effectively.

References

Anthropic (2024) ‘Multi-agent research system’. Available at: **https://www.anthropic.com/engineering/multi-agent-research-system** (Accessed: 3 August 2026).

Domo (2026) ‘AI agent examples’. Available at: **https://www.domo.com/learn/article/ai-agent-examples** (Accessed: 2 August 2026).

Massachusetts Institute of Technology Media Lab (2025) ‘On the limits of agency in agent-based models’. Available at: **https://www.media.mit.edu/publications/on-the-limits-of-agency-in-agent-based-models/** (Accessed: 2 August 2026).

Nature Index (n.d.) ‘Agent-based modelling and simulation in complex systems’. Available at: **https://www.nature.com/nature-index/topics/l4/agent-based-modeling-and-simulation-in-complex-systems** (Accessed: 2 August 2026).

Sahin, S. (2024) ‘Agent-based AI systems: What is it and how it works’. *Medium*. Available at: **https://medium.com/@sahin.samia/agent-based-ai-systems-what-is-it-and-how-it-works-dd6b76c19352** (Accessed: 3 August 2026).

E3 Magazine (2026) ‘GenAI impact study: Seven principles of agent-based software development’. Available at: **https://e3mag.com/en/genai-impact-study-seven-principles-of-agent-based-software-development/** (Accessed: 3 August 2026).

Peer Response

by Ali Nahhas - Thursday, 6 August 2026, 1:44 PM

Your post explains how RAG, fine-tuning and multi-agent collaboration have expanded what AI systems can do. However, it risks treating every AI tool as an agent. That distinction matters because better answers do not automatically mean greater autonomy.

RAG retrieves external information to improve generation, while an autonomous agent also needs mechanisms for planning, memory, action and feedback (Zhao et al., 2026; Wang et al., 2024). A chatbot connected to a database may therefore be useful without independently pursuing goals. The same issue applies to agent-based modelling. Dehkordi et al. (2023) describe it as a method for studying system-level patterns emerging from individual interactions. Its purpose is simulation, not completing organisational tasks in the real world.

This weakens the Wealthfront example. Automated portfolio allocation may be sophisticated, but rules-based personalisation is not necessarily agency. Calling it an agent could exaggerate how much independent judgement the system has.

Organisations should first identify whether a system retrieves information, simulates behaviour, automates rules or acts autonomously. These categories can overlap, but they should not be treated as interchangeable. Otherwise, businesses may purchase conventional automation under an “agentic” label and expect reasoning, adaptability and collaboration that the system was never designed to provide reliably.

References

Dehkordi, M.A.E., Lechner, J., Ghorbani, A., Nikolic, I., Chappin, E. and Herder, P. (2023) ‘Using machine learning for agent specifications in agent-based models and simulations: A critical review and guidelines’, *Journal of Artificial Societies and Social Simulation*, 26(1), article 9.

Wang, L., Ma, C., Feng, X., Zhang, Z., Yang, H., Zhang, J., Chen, Z., Tang, J., Chen, X., Lin, Y., Zhao, W.X., Wei, Z. and Wen, J. (2024) ‘A survey on large language model based autonomous agents’, *Frontiers of Computer Science*, 18, article 186345.

Zhao, P., Zhang, H., Yu, Q., Wang, Z., Geng, Y., Fu, F., Yang, L., Zhang, W., Jiang, J. and Cui, B. (2026) ‘Retrieval-augmented generation for AI-generated content: A survey’, *Data Science and Engineering*, 11, pp. 1–29.

In reply to Yasmin Almeida Ferrera

Re: Initial Post

by Yasmin Almeida Ferrera - Wednesday, 12 August 2026, 10:55 AM

Summary Post

In my original post, I discussed how the development of technologies such as RAG, fine-tuning, external tools and multi-agent collaboration has expanded the capabilities of AI systems and increased their potential value for organisations. However, Ali's response made me reconsider how broadly I used the term "AI agent". I agree that improving an AI system does not necessarily mean that it has become an autonomous agent.

For example, RAG can provide an LLM with access to external and more up-to-date information, but this alone does not give the system the ability to independently plan and act towards a goal. Autonomous agents require additional capabilities such as planning, memory, tool use and feedback (Wang et al., 2024). I also agree with Ali's distinction between agent-based modelling and agent-based systems. Agent-based modelling is mainly used to simulate interactions between individual agents and understand the wider patterns that emerge from them (Dehkordi et al., 2023).

This also made me reconsider the Wealthfront example from my original post. Although automated investment platforms can make decisions based on user preferences and market information, this does not automatically make them autonomous agents. Their classification depends on how independently they can plan, adapt and take action rather than simply following predefined rules.

Overall, Ali's response helped me refine my argument. Organisations should be careful not to use "AI agent" as a general label for any advanced AI system. As agentic AI becomes more common, at what point does an AI system move from sophisticated automation to genuine autonomy?

References

Dehkordi, M.A.E. et al. (2023) 'Using machine learning for agent specifications in agent-based models and simulations: A critical review and guidelines', Journal of Artificial Societies and Social Simulation, 26(1), article 9.

Wang, L. et al. (2024) 'A survey on large language model based autonomous agents', Frontiers of Computer Science, 18, article 186345.